

Sudhanva Manjunath Athreya

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RESEARCH INTERESTS

Foundational Machine Learning, Reinforcement Learning, Recommender Systems, Computational Topology, Sequential Decision-Making

SUMMARY

I'm a lifelong student who's highly interested in technology and mathematics.

EDUCATION

2020 - 2024	Bachelor's in Computer Science at Reva University, Bangalore	(GPA: 8.9/10)
2024 - Present	PhD in Computer Science at University of Utah, Salt Lake City	(GPA: 3.8/4)

WORK EXPERIENCE

Scientific Computing and Imaging (SCI) Institute - Student Researcher Jan 2025 - Present

- Working under Prof. Paul Rosen to interpret deep learning models using techniques from computational topology.
- Developed HOLE (Homological Observation of Latent Embeddings), a method for analyzing neural networks through persistent homology. Paper published on arXiv (arXiv:2512.07988) and under review.
- Used persistent homology to analyze the structure of activation spaces in models like Vision Transformers (ViTs) and ResNets.
- Created multiple visualization techniques based on Sankey diagrams, Dendrograms, and Force-directed graphs to visualize the topological features.
- Presented a poster at SCI-X 2025, titled "Interpreting Deep Learning Models using Persistent Homology".
- Developed a Python package and dashboard to make these methods a standard tool for model-agnostic explainability and interpretability.

Medical Machine Intelligence Lab - Graduate Research Assistant Aug 2025 - Present

1. Developed a Mixture-of-Experts (MoE) based foundational model to efficiently predict patient timelines from EHR data.
2. Co-authored and presented a poster at AI Summit 2025 with Matt Christenson and Prof. Warren Petline, titled "Efficient EHR Foundational Models: A Mixture-of-Experts Approach for Patient Timeline Prediction".
3. Presented a poster and lightning talk at Utah AI Convergence 2026, titled "An EHR foundation model learns a clinical geometry", showing that next-token pretraining on MIMIC-IV alone recovers clinical structure in the embedding space: drug classes sit nearest the diseases they treat (83%, $p < 0.001$), lab and BMI values form ordered low-to-high axes ($|\rho|$ 0.89–0.98), and elapsed-time tokens are spaced logarithmically (R^2 0.85 vs 0.18).

MTN Biometrics - Machine Learning Intern Jun 2025 - Aug 2025

- Worked on "Summit," an ETL pipeline for processing and cleaning high-volume biometric data streams.
- Designed interactive visualization components for real-time monitoring and analysis of biometric data.
- Designed and deployed a RESTful API for the Summit platform, enabling seamless data and pipeline access for downstream applications.

1. Working under Prof. Warren Pettine to build EHR and Genomic Foundational models.
2. Reproduced and benchmarked EHR foundation models on MIMIC-IV FHIR using the EHRMamba and Odyssey toolkits, evaluating against XGBoost, LSTM, BigBird and Multibird baselines on mortality, length-of-stay and readmission prediction.
3. Built the tokenization pipeline mapping FHIR resources (diagnoses, procedures, medications, lab events) into time-ordered clinical token streams with visit and time-interval tokens, supporting both masked-language-modelling and next-token pretraining objectives.
4. Ran discharge and length-of-stay prediction experiments with smaller models (XGBoost, TabPFN, Med-BERT, ClinicalBERT) on MIMIC-IV, focusing on cohort construction and feature modelling.
5. Scheduled distributed training jobs on the lab's SLURM HPC cluster and profiled model latency, memory, FLOPs and parameter counts with the PyTorch Profiler.
6. Surveyed the EHR foundation-model literature (ETHOS, Clinical Language Models, and Foundational Models for EHRs) to guide the lab's architecture and evaluation choices.

Siemens - Computer Vision Research Intern

Jun 2023 - Jun 2024

- EdgeAI : A research project where the bridged domains of ML and Compilers were explored.
 1. Created a unified file-format for Siemens for model storage and faster loading using protobuf.
 2. Created pipelines incorporating techniques such as chunking, memory-mapping & lazy-loading to reduce memory-consumption during inference.
 3. Created inference and quantization pipelines.
- AI4Safety Edge : A Siemens monitoring application for edge ecosystems.
 1. Setup [Deepstream](#) pipeline and components on Jetson Orin Nano and Siemens Edge cameras.
 2. Coupled a rule engine which recorded violations.
 3. Quantized detection and zero-shot models and integrated it into the Deepstream pipeline.
 4. Created Gstreamer plugins to integrate Owl-ViT into Deepstream.
 5. Created a scene understanding pipeline using LVMs (CLIP)
 6. Explored VisualSLAM, ORB-SLAM, SfM, Photogrammetry, COLMAP, NeRF, Gaussian Splatting for 3d-Scene reconstruction.
- AI4Safety : A Siemens Surveillance product for factories and logistics environments.
 1. Dataset curation, training & evaluation.
 2. Created custom Discrete Correlation Filter-based tracker and integrated it with the Deepstream pipeline.
 3. Model optimization - Pruning, Quantization & Palletization
 4. Data de-duplication, Denoising & Augmentation pipelines
 5. Actively contributed in the creation of a rule-engine
 6. Created Homography pipeline to perform object tracking in a BEV (Birds-Eye-View) plane.
 7. Created and tested Deepstream plugins
- SpeCT Analysis : Scintillation Crystals defect analysis framework.
 1. Created a MLops dashboard, API, datastore and containerized the application.
- Domains : Computer Vision, Deep Learning, Software development
- Technologies : OpenCV, Nvidia Deepstream, Nvidia TensorRT, YOLO, Pytorch, C++, Docker, Kafka

Tata Elxsi - NLP Intern

Jul 2022 - Oct 2022

- LMS Recommendation System : Recommendation system for Tata Elxsi LMS platform
 1. Used [SLiM \(Sparse Linear Methods\)](#) for collaborative-filtering.
 2. Creation of Responsive React UI for the recommendation page.
 3. Creation of Flask API, mongoDB user database integration & containerization.
- VideoPeek
 1. Created a search page similar to [Google Key moments](#) for LMS platform
 2. Made use of Haystack framework & Milvus DB to setup semantic search pipeline
 3. Setup Whisper pipeline to transcribe new courses & add it to the vector DB
- Domains : NLP, Software development
- Technologies : Python, Haystack, Pytorch, ReactJS, MongoDB

- Revival of Sharada scripture (funded by DST under SHRI)
 1. Ongoing project which aims to revive Sharada scripture used in the Kashmir region by the Kashmiri Pandits to write Sanskrit and Kashmiri Manuscripts
 2. Creation of HTR system using CTPN-CRNN & CTC loss.
 3. Creation of custom customized augmentation pipeline specific to handwritten text.
 4. Transliteration of Sharada Scripture.
 5. Creation of a web portal for awareness.
- Domains : Computer Vision, NLP, Deep Learning
- Technologies : OpenCV, Tesseract, YOLO, Pinecone, Tensorflow, Pytorch, Pillow

PROJECTS

DS-Checker

Domain : NLP, Software Development

Created as a year-end Project (1st year Bachelors). It calculates the similarity between documents and displays it. It was mainly created to check the plagiarism among the project papers submitted by the students. Initially, the documents were embedded using TF-IDF values and then passed through a cosine similarity function to determine their similarity. The system allowed users to upload documents through a PHP-based website. Later, the project was migrated to utilize Longformer embeddings, and the user interface was revamped using streamlit.

Big Lens

Domain : Computer Vision, Deep Learning, Web Development

Created as a year-end Project (2nd year Bachelors). This open-source reverse-image search engine was designed to emulate popular services like Google, Yandex, Bing, and TinEye. Initially, it made use of APIs of previously mentioned services. Eventually, the project was transitioned to employ Content-Based Image Retrieval (CBIR), similar to the approach used by Yandex. A VGG19 model was for feature extraction and streamlit interface could be used to interact with the model. hsnwlib along with sqlite was integrated which allows for a fast nearest neighbor search.

Kailāsaḥ

Domain : NLP, Deep Learning, Web Development

Created as a project for the Sambhasha Sanskrit Conference. Kailāsaḥ is a website which performs Sanskrit Sandhi split. Sandhi's are similar to contraction in English, used extensively in Sanskrit to speak quickly and smoothly by condensing large sentences. It is one of the projects done for DST, Govt. of India. It makes use of a CRNN model to perform word splitting. The project has a streamlit interface. It also provides with the required contextual meaning of base words.

Hestia

Domain : NLP, Software Development

Hestia is an extension for your browser which allows you to search your bookmarks and bookmark folders. The extension can be downloaded from the Firefox add-on store. The extension performs a search on the bookmarks to return the results

PUBLICATIONS

“Using Deep Learning Techniques to Evaluate Lung Cancer Using CT Images” (Jan. 2022). In: *SN Computer Science*.

“Machine Learning Techniques to Detect DDoS Attacks in IoT's, SDN's: A Comprehensive Overview” (June 2023). In: *International Journal of Human Computations and Intelligence*.

Athreya, Sudhanva Manjunath and Paul Rosen (Dec. 2025). “HOLE: Homological Observation of Latent Embeddings for Neural Network Interpretability”. In: *IEEE Transactions on Visualization and Computer Graphics (TVCG)*. Under review.

Athreya, Sudhanva Manjunath and Sai Phani Kumar Malladi (Aug. 2026). “More Accurate, Less Human: Gestalt Grouping in Vision Models”. In: *VISxVision Workshop, IEEE VIS*. Accepted.

EVENTS

Paninian Grammar and its Applications	Demonstrated Kailāsaḥ tool (14th Feb 2023)
SIEMENS Shift healthcare hackathon finalist (Nov 2022)	Led a team where we built a hospital bed queueing and allocation linear model. We also build a React based dashboard to manage the hospital. (18th Nov 2022)
Generative AI and Cybersecurity FDP Seminar (Jul 2023)	Gave a talk on Zero-shot learning in Computer Vision Applications during the 5-day Faculty Development Program.

HONORS & GRANTS

DST SHRI 2022

- DSP/TDT/SHRI-14/2021
- Date: 13 December 2021
- An Artificial Intelligence based system for the preservation, restoration and translation of the prominent Sharda literature of Jammu and Kashmir
- Working under Dr. Nimrita Koul on the project funded by the Govt. Of India, Department of Science and Technology(DST) - Science and Heritage Research Initiative (SHRI)

OTHERS

JLPT N5 2022 - Japanese Language Proficiency Test